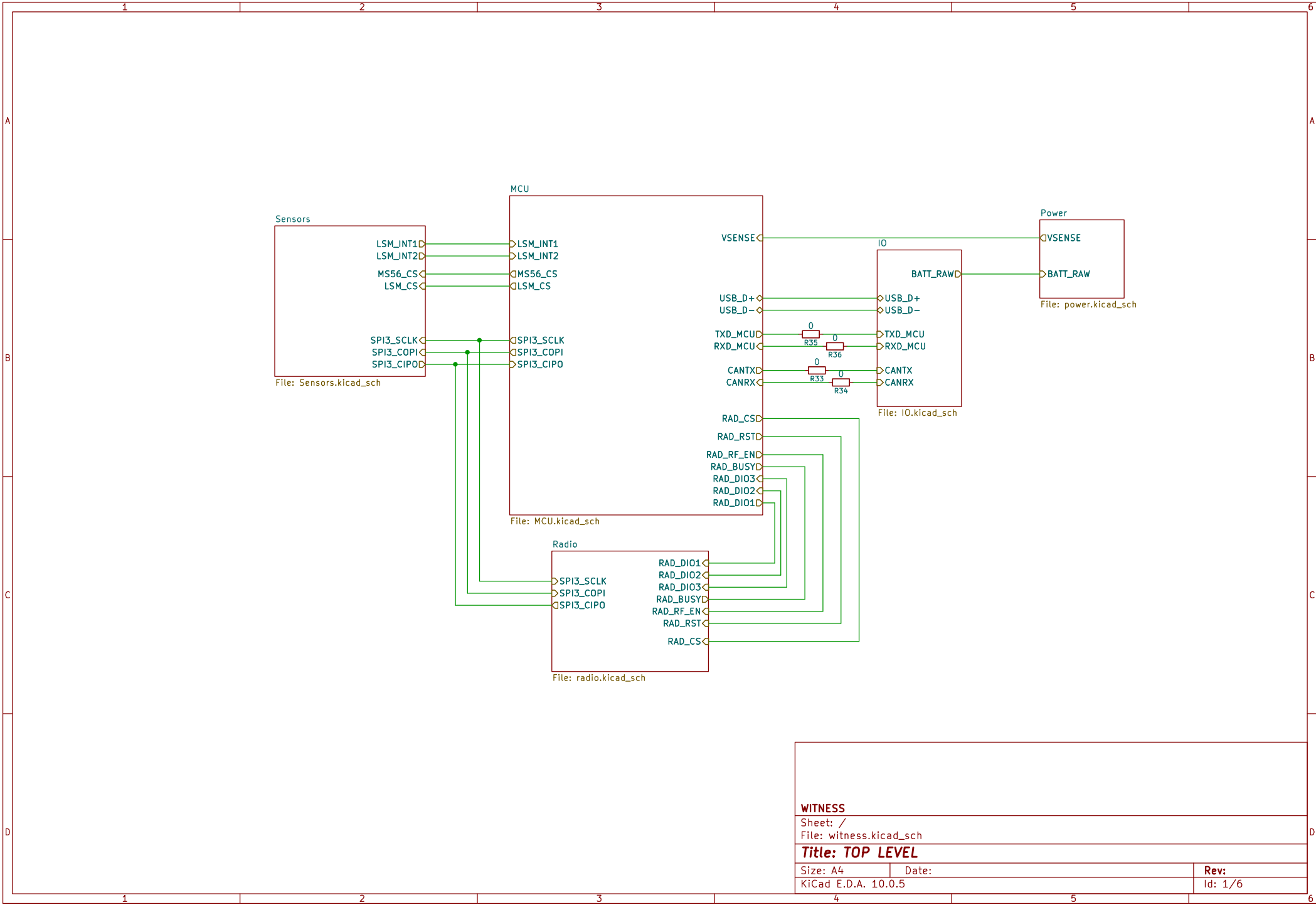




**WITNESS**

Sheet: /  
File: witness.kicad\_sch

**Title: TOP LEVEL**

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Table 3-1. Default Configuration of Strapping Pins

| Strapping Pin | Default Configuration | Bit Value |
|---------------|-----------------------|-----------|
| GPIO0         | Weak pull-up          | 1         |
| GPIO3         | Floating              | —         |
| GPIO45        | Weak pull-down        | 0         |
| GPIO46        | Weak pull-down        | 0         |

### 3.1 Chip Boot Mode Control

GPIO0 and GPIO46 control the boot mode after the reset is released. See Table 3-3 Chip Boot Mode Control.

Table 3-3. Chip Boot Mode Control

| Boot Mode                             | GPIO0 | GPIO46    |
|---------------------------------------|-------|-----------|
| SPI boot mode                         | 1     | Any value |
| Joint download boot mode <sup>2</sup> | 0     | 0         |

<sup>1</sup> Bold marks the default value and configuration.  
<sup>2</sup> Joint Download Boot mode supports the following download methods:

- USB Download Boot:
  - ~ USB-Serial\_JTAG Download Boot
  - ~ USB-OTIO Download Boot
- UART Download Boot

Figure 1-1. ESP32-S3 Series Nomenclature

Table 1-1. ESP32-S3 Series Comparison

| Part Number <sup>1</sup> | In-Package Flash <sup>2</sup> | In-Package PSRAM | Ambient Temp. <sup>3</sup> | VDD_SPI Voltage <sup>4</sup> | Chip Version |
|--------------------------|-------------------------------|------------------|----------------------------|------------------------------|--------------|
| ESP32-S3                 | —                             | —                | ~40 ~ 105 °C               | 3.3 V/1.8 V                  | v0.1/v0.2    |
| ESP32-S3FN8              | 8 MB (Quad SPI) <sup>5</sup>  | —                | ~40 ~ 85 °C                | 3.3 V                        | v0.1/v0.2    |

The VDD\_SPI voltage can be:

- (Default) 3.3 V supplied by VDD3P3\_RTC via R<sub>PU</sub>
- 1.8V supplied by the Flash Voltage Regulator

The voltage is determined by GPIO0, SPI1\_FORCE[0], and GPIO45, SPI1\_FORCE[1].

Table 3-4. VDD\_SPI Voltage Control

| VDD_SPI power source <sup>1</sup> | Voltage | GPIO0 | GPIO45 | SPI1_FORCE[0] | SPI1_FORCE[1] |
|-----------------------------------|---------|-------|--------|---------------|---------------|
| VDD3P3_RTC via R <sub>PU</sub>    | 3.3 V   | 0     | 0      | Ignored       | Ignored       |
| VDD3P3_RTC via R <sub>PU</sub>    | 3.3 V   | 1     | 0      | Ignored       | 1             |
| Flash Voltage Regulator           | 1.8 V   | 0     | 1      | Ignored       | Ignored       |
| Flash Voltage Regulator           | 1.8 V   | 1     | 1      | Ignored       | Ignored       |

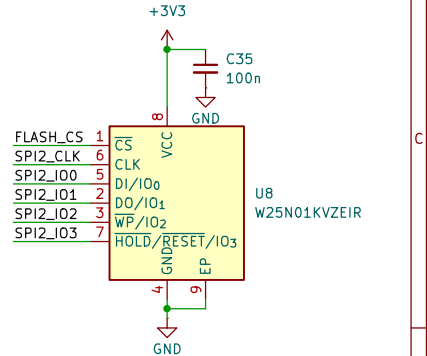
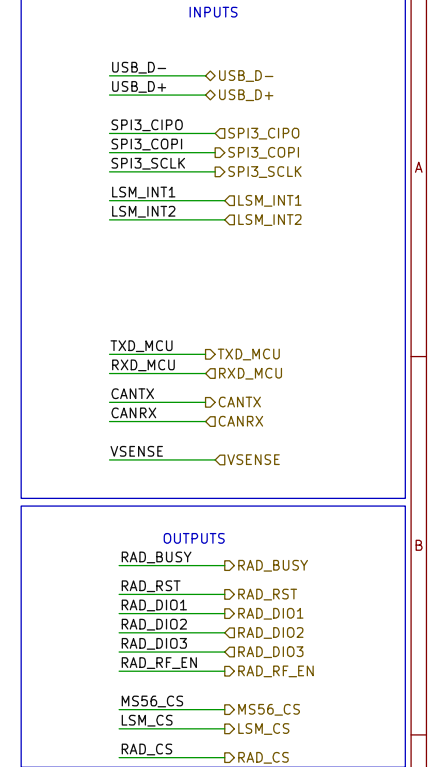
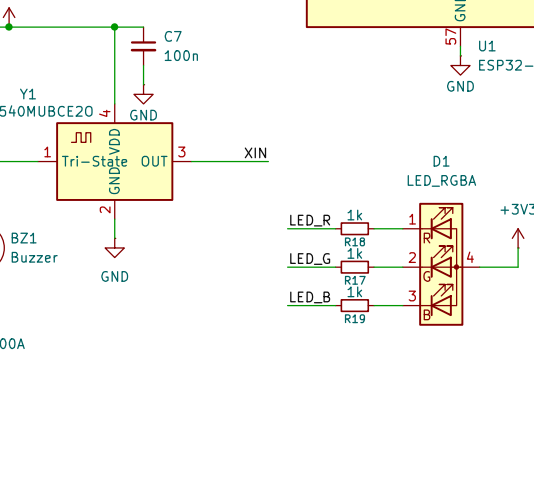
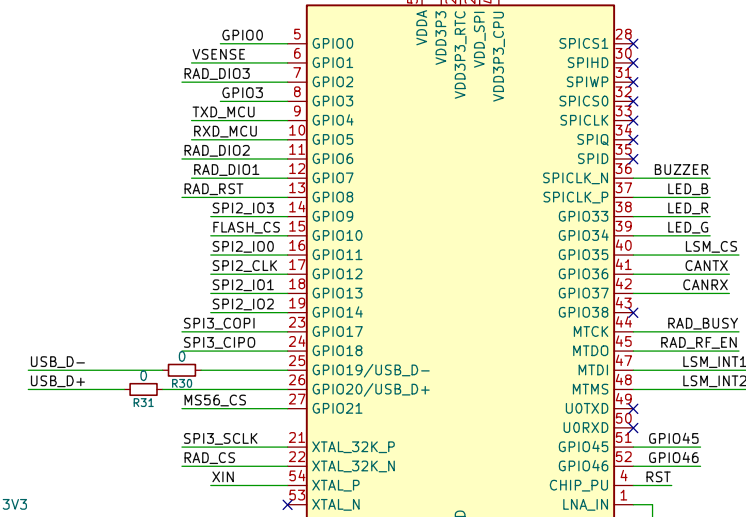
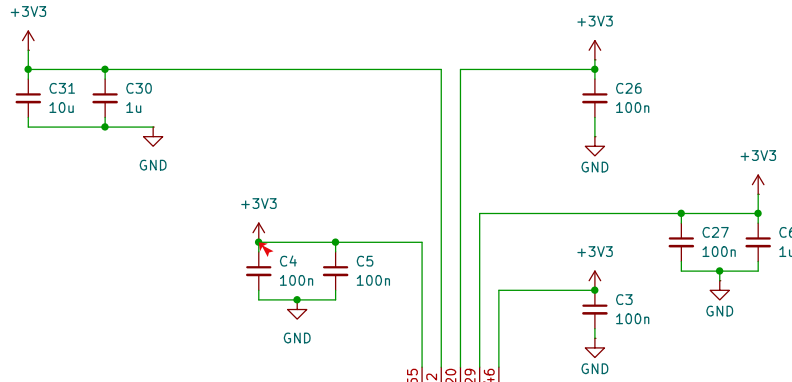
<sup>1</sup> Bold marks the default value and configuration.  
<sup>2</sup> 0: Pin connected to MTDO, MTMS, MTMS, and MTDO.

<sup>3</sup> See Section 3.3.5 Power Scheme.

Table 3-5. JTAG Signal Source Control

| JTAG Signal Source         | GPIO0 | GPIO45 | JTAG_ENABLE | JTAG_DISABLE | JTAG_ENABLE | JTAG_DISABLE |
|----------------------------|-------|--------|-------------|--------------|-------------|--------------|
| USB-Serial/JTAG Controller | 0     | 0      | 0           | Ignored      | 0           | 1            |
| JTAG pin <sup>2</sup>      | 1     | 0      | 0           | Ignored      | 1           | 0            |
| JTAG pin <sup>2</sup>      | 0     | 0      | 1           | Ignored      | 0           | 0            |
| JTAG pin <sup>2</sup>      | 1     | 1      | 1           | Ignored      | 1           | 1            |

<sup>1</sup> Bold marks the default value and configuration.  
<sup>2</sup> 0: Pin connected to MTDO, MTMS, MTMS, and MTDO.



### WITNESS

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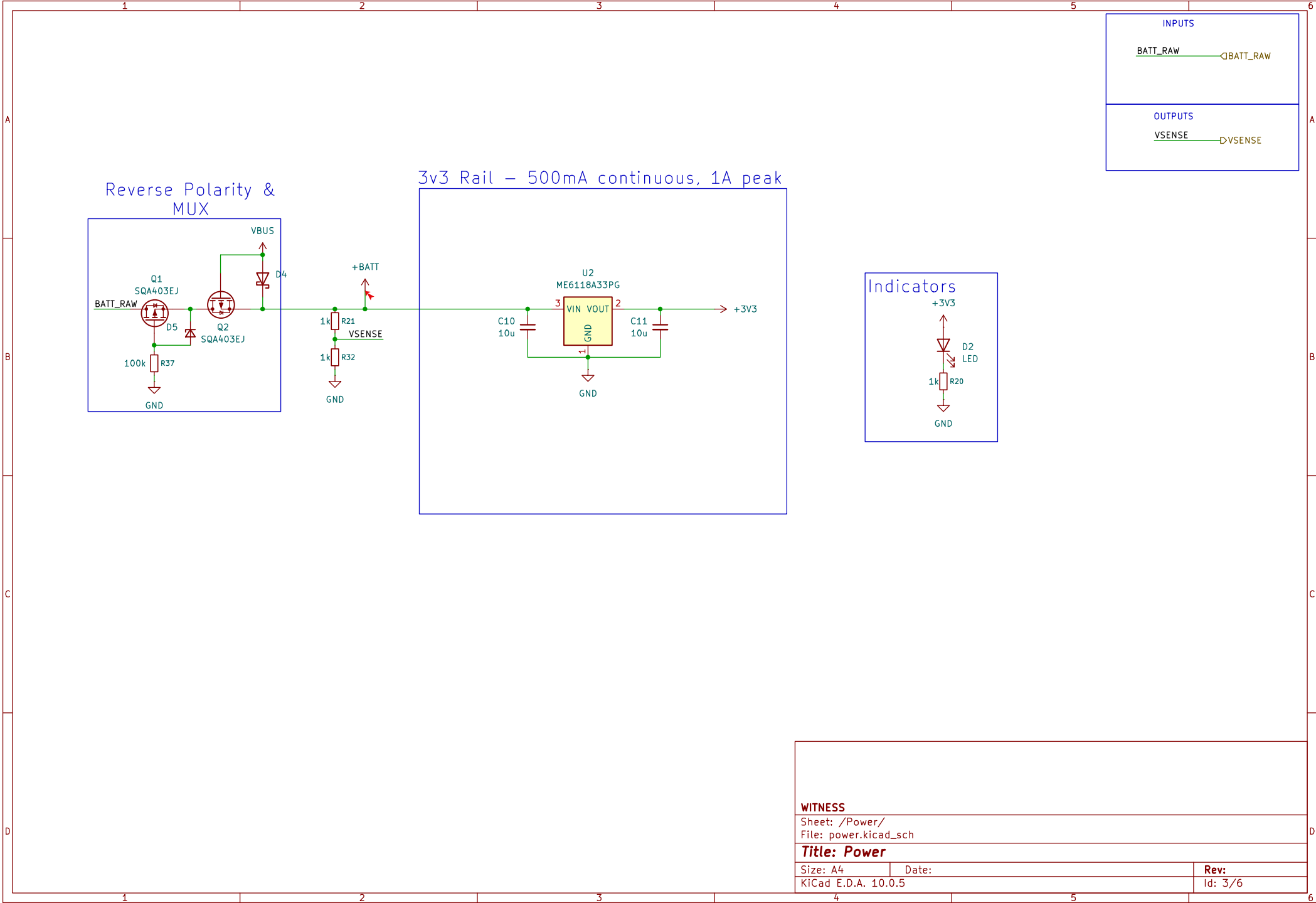
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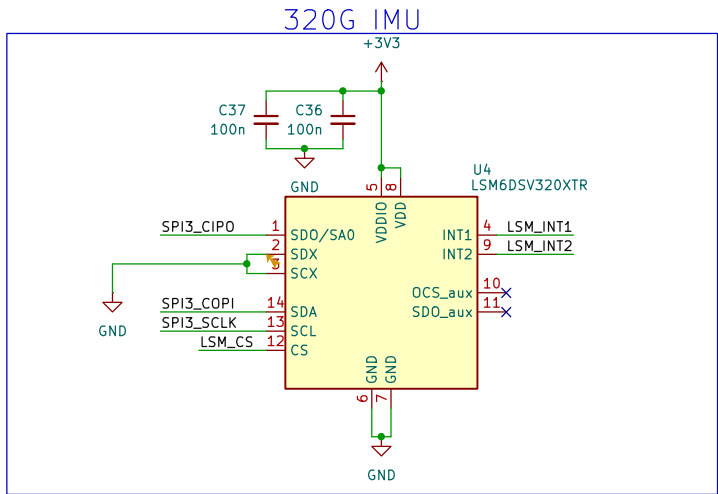
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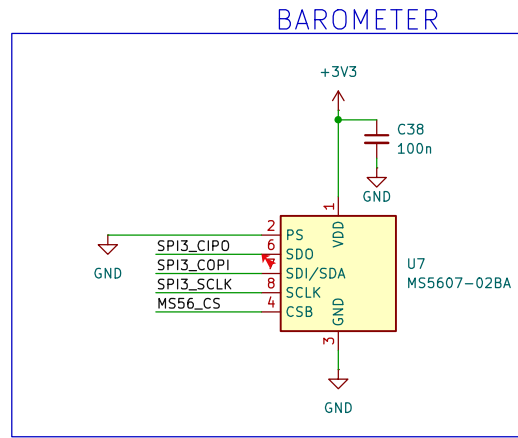
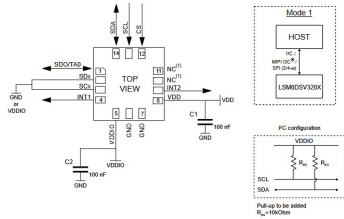
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LSM6DSV320X electrical connections in mode 1

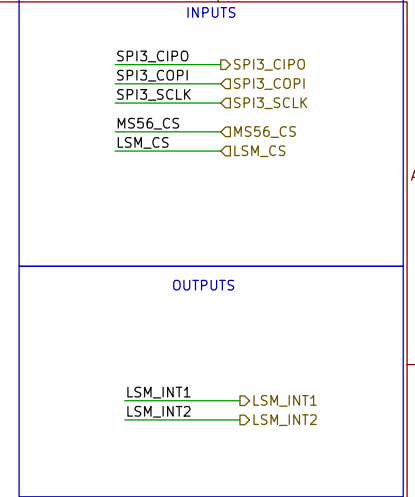
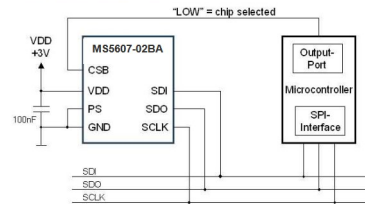
Figure 27. LSM6DSV320X electrical connections in mode 1



#### APPLICATION CIRCUIT

The MS5607-02BA is a circuit that can be used in conjunction with a microcontroller in mobile altimeter applications. It is designed for low-voltage systems with a supply voltage of 3 V.

#### SPI protocol communication



#### WITNESS

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File: Sensors.kicad\_sch

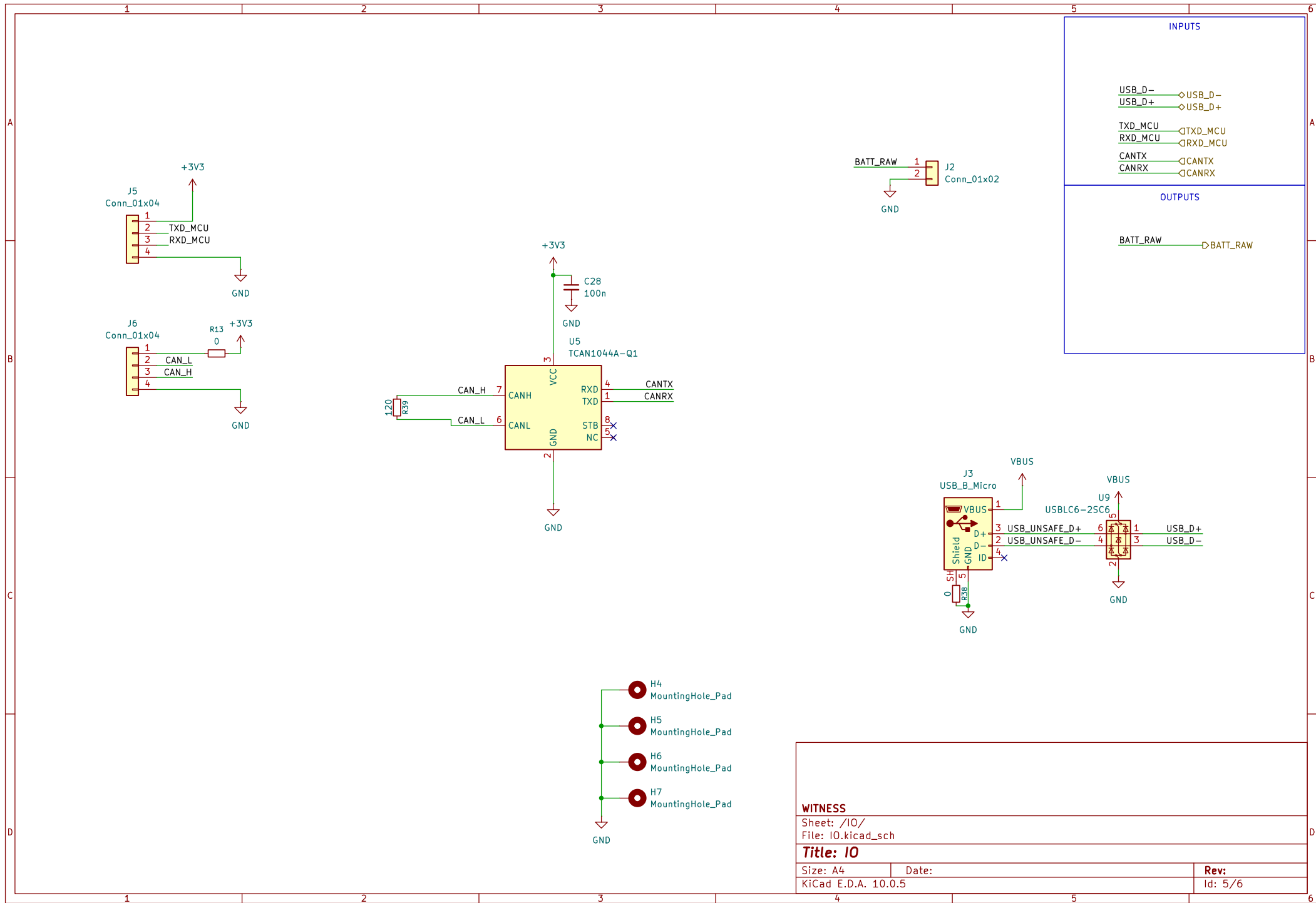
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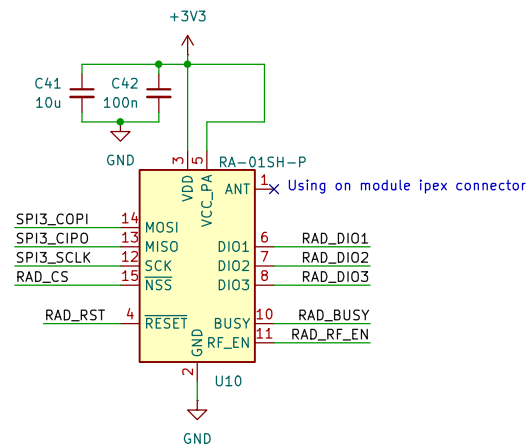
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| INPUTS    |             |
|-----------|-------------|
| SPI3_COPI | → SPI3_COPI |
| SPI3_CIPO | → SPI3_CIPO |
| SPI3_SCLK | → SPI3_SCLK |
| RAD_CS    | → RAD_CS    |
| RAD_RST   | → RAD_RST   |
| RAD_DIO1  | → RAD_DIO1  |
| RAD_DIO2  | → RAD_DIO2  |
| RAD_DIO3  | → RAD_DIO3  |
| RAD_RF_EN | → RAD_RF_EN |

| OUTPUTS  |            |
|----------|------------|
| RAD_BUSY | → RAD_BUSY |



Ra-01SH-P Specification V1.0.0

## (2) Typical application circuit

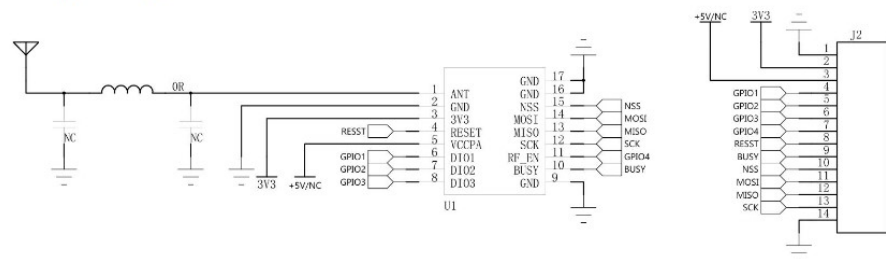


Figure 7 Typical application circuit

### WITNESS

Sheet: /Radio/  
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### Title: RADIO

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